

Meson Octet

$3: u, d, s$

$SU(3)_{\text{Flavor}}$

$\bar{3}: \bar{u}, \bar{d}, \bar{s}$

Static quark model

$$3 \otimes \bar{3} = 1 \oplus 8$$

$$\text{Meson} \equiv q_i \bar{q}_j$$

$I_3, \text{Strangeness}$

$\begin{pmatrix} u \\ d \end{pmatrix} I = 1/2 \text{ doublet} \quad s: I = 0 \text{ singlet.}$

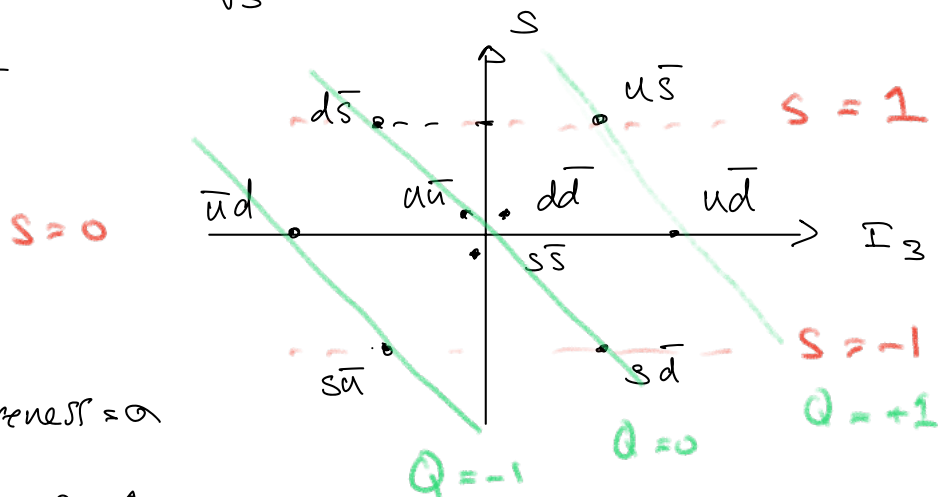
Strangeness: ± 0 for u, d . $S = -1$ for s .

$3 \times 3 = 9$ possible states $q_i \bar{q}_j$.

1 : Flavor singlet

$$\frac{1}{\sqrt{3}} (u\bar{u} + d\bar{d} + s\bar{s}) \quad S=0, I=0, I_3=0$$

8 : flavor octet



Flavor octet:

2 states total strangeness ± 1

$I_3 = 0$ total. $I = 0, 1$

$$\psi_{F, I} = \psi_{8, 1, I_3=0} = \frac{1}{\sqrt{2}} (u\bar{u} - d\bar{d})$$

↙
↘

flavor
 I resp. n

$$\psi_{8, I=0, I_3=0} = \frac{1}{\sqrt{6}} (u\bar{u} + d\bar{d} - 2s\bar{s})$$

Singlet $\psi_{1, I=0, I_3=0} = \frac{1}{\sqrt{3}} (u\bar{u} + d\bar{d} + s\bar{s})$

$$m_K \approx 500 \text{ MeV}$$

$$m_{\pi^\pm} \approx 140 \text{ MeV}$$

$$m_{\pi^0} \approx 140 \text{ MeV}$$

$I = 1$, triplet of isospin.

$$\begin{cases} |\pi^0\rangle = \frac{1}{\sqrt{2}}(u\bar{u} - d\bar{d}) \\ |\pi^+\rangle = |u\bar{d}\rangle \\ |\pi^-\rangle = |d\bar{u}\rangle \end{cases} \quad \text{approximately degenerate in mass.}$$

$$\begin{cases} |K^-\rangle = |s\bar{u}\rangle & I_3 = -1/2 \\ |K^0\rangle = |s\bar{d}\rangle & I_3 = +1/2 \end{cases} \quad S = -1$$

$$8 = \text{hexagon with } \pi^0 \text{ and } ? \text{ inside} + \psi_{1,0}$$

if u, d, s had exactly same mass $\Rightarrow K, \pi$ have same mass.

instead $M_K > M_\pi \Rightarrow M_s > M_{u,d}$, $m_u \approx m_d$
 $SU(3)_{\text{Flavor}}$ is a very good approximate symm.

2 states left with $I_3 = 0$

$$\psi_{8,1,I_3=0} \quad \psi_{1,1,I=0,I_3=0} \longleftrightarrow \eta', \eta^0$$

possible states from group theory. physical states observed.

$$\begin{cases} f = \psi_{8,0} \cos \theta_{\text{mix}} - \psi_{1,0} \sin \theta_{\text{mix}} \\ f' = \psi_{8,0} \sin \theta_{\text{mix}} + \psi_{1,0} \cos \theta_{\text{mix}} \end{cases} \quad \theta_{\text{mix}}: \text{light meson mixing angle}$$

$$f, f' \text{ linear combination of } \left. \begin{array}{l} I=0, I_3=0, F=8 \\ I=0, I_3=0, F=1 \end{array} \right\} \begin{array}{l} \psi_{8,0} \\ \psi_{1,0} \end{array}$$

F, I

η, η' are comb. of $\psi_{8,0}$ and $\psi_{1,0}$

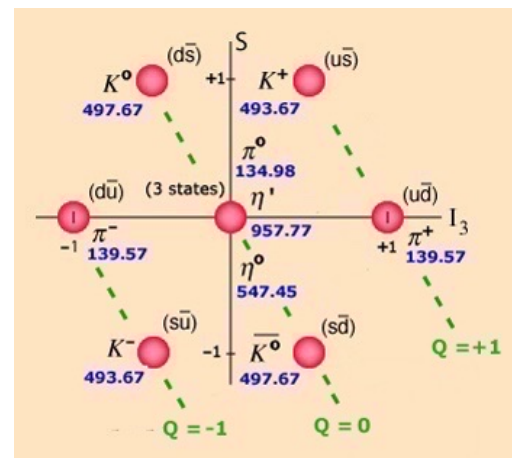
$$M = q_i \bar{q}_j \quad S = 0, 1$$

$q_i, \bar{q}_j: S = 1/2 \Rightarrow$

$\uparrow \downarrow$
 $\sim$
 $S=0$

$\uparrow \uparrow \quad \downarrow \downarrow$
 $\sim$
 $S=1$

Meson = $1 \oplus 8$
 vector meson $S=1$
 pseudo-scalar meson $S=0$

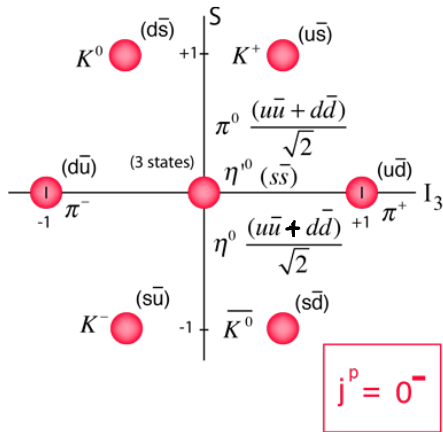


pseudo-scalar octet:

$$K^{\pm}, K^0, \bar{K}^0, \bar{u}^{\pm}, \bar{u}^0, 1, 1'$$

$$S=0$$

$$\psi_{\text{meson}} = \psi_{\text{space}} \psi_{\text{flavor}} \psi_{\text{spin}}$$



$$S=0$$

$$L=0 \text{ (ground state)} \Rightarrow$$

$$J=0$$

$$P_{q\bar{q}} = (-1)(-1)^L = -1$$

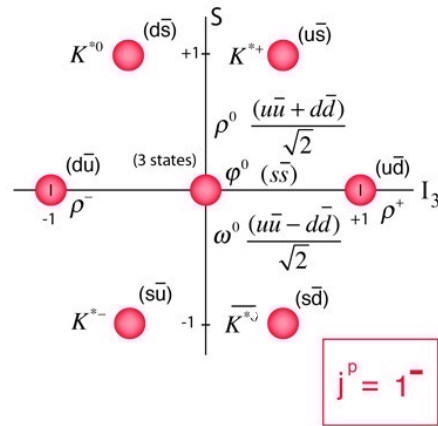
$q\bar{q}$ from QFT

vector-meson octet

$$K^{* \pm}, K^{*0}, \bar{K}^{*0}, \rho^{\pm}, \rho^0, \phi, \omega$$

$$S=1$$

vector octet



$$S=1$$

$$L=0 \text{ (ground state)} \Rightarrow J=1$$

$$P = (-1)(-1)^L = -1$$

$\swarrow$
from QFT

$$\left. \begin{aligned} \pi^0(140) &\approx \psi_{8,1}^{ps} = (u\bar{u} - d\bar{d})/\sqrt{2} \\ \eta(550) &= \psi_{8,0}^{ps} \cos\theta_{ps} - \psi_1^{ps} \sin\theta_{ps} \\ \eta'(960) &= \psi_{8,0}^{ps} \sin\theta_{ps} + \psi_1^{ps} \cos\theta_{ps} \end{aligned} \right\} \begin{aligned} J^P &= 0^-, \\ \theta_{ps} &\approx -25^\circ \end{aligned}$$

pseudo-scalar mixing angle

include some $s\bar{s}$ content

$$I_3=0, S=0$$

$$\begin{aligned} \psi_{8,1} &= (u\bar{u} - d\bar{d})/\sqrt{2} \\ \psi_{8,0} &= (u\bar{u} + d\bar{d} - 2s\bar{s})/\sqrt{6} \\ \psi_1 &= (u\bar{u} + d\bar{d} + s\bar{s})/\sqrt{3} \end{aligned}$$

$$\left. \begin{aligned} \rho^0(770) &\approx \psi_{8,1}^v = (u\bar{u} - d\bar{d})/\sqrt{2} \\ \phi(1020) &= \psi_{8,0}^v \cos\theta_v - \psi_1^v \sin\theta_v \approx s\bar{s} \\ \omega(780) &= \psi_{8,0}^v \sin\theta_v + \psi_1^v \cos\theta_v \approx (u\bar{u} + d\bar{d})/\sqrt{2} \end{aligned} \right\} \begin{aligned} J^P &= 1^-, \\ \theta_{\text{vect}} &\approx 36^\circ \end{aligned}$$

vector meson mixing angle

$$\theta \approx 30^\circ \quad \cos 30^\circ = \frac{\sqrt{3}}{2}$$

$$\sin 30^\circ = \frac{1}{2}$$

$M = q_i q_j \Rightarrow$ compute mass of states with different L, S

$$U \approx -\frac{\alpha_S}{r} + Kr$$

$n^{2s+1}\ell_J$	J^{PC}	$I = 1$ $u\bar{d}, \bar{u}d,$ $\frac{1}{\sqrt{2}}(d\bar{d} - u\bar{u})$	$I = \frac{1}{2}$ $u\bar{s}, d\bar{s};$ $\bar{d}s, \bar{u}s$	$I = 0$ f	$I = 0$ f'
1^1S_0	0^{-+}	π	K	η	$\eta'(958)$
1^3S_1	1^{--}	$\rho(770)$	$K^*(892)$	$\phi(1020)$	$\omega(782)$
1^1P_1	1^{+-}	$b_1(1235)$	K_{1B}^a	$h_1(1415)$	$h_1(1170)$
1^3P_0	0^{++}	$a_0(1450)$	$K_0^*(1430)$	$f_0(1710)$	$f_0(1370)$
1^3P_1	1^{++}	$a_1(1260)$	K_{1A}^a	$f_1(1420)$	$f_1(1285)$
1^3P_2	2^{++}	$a_2(1320)$	$K_2^*(1430)$	$f_2'(1525)$	$f_2(1270)$
1^1D_2	2^{-+}	$\pi_2(1670)$	$K_2(1770)^a$	$\eta_2(1870)$	$\eta_2(1645)$
1^3D_1	1^{--}	$\rho(1700)$	$K^*(1680)^b$	$\phi(2170)^d$	$\omega(1650)$
1^3D_2	2^{--}		$K_2(1820)^a$		
1^3D_3	3^{--}	$\rho_3(1690)$	$K_3^*(1780)$	$\phi_3(1850)$	$\omega_3(1670)$
1^3F_4	4^{++}	$a_4(1970)$	$K_4^*(2045)$	$f_4(2300)$	$f_4(2050)$
1^3G_5	5^{--}	$\rho_5(2350)$	$K_5^*(2380)$		
2^1S_0	0^{-+}	$\pi(1300)$	$K(1460)$	$\eta(1475)^c$	$\eta(1295)$
2^3S_1	1^{--}	$\rho(1450)$	$K^*(1410)^b$	$\phi(1680)$	$\omega(1420)$
2^3P_1	1^{++}	$a_1(1640)$			
2^3P_2	2^{++}	$a_2(1700)$	$K_2^*(1980)$	$f_2(1950)$	$f_2(1640)$

$$f = \psi_8 \cos \theta - \psi_1 \sin \theta$$

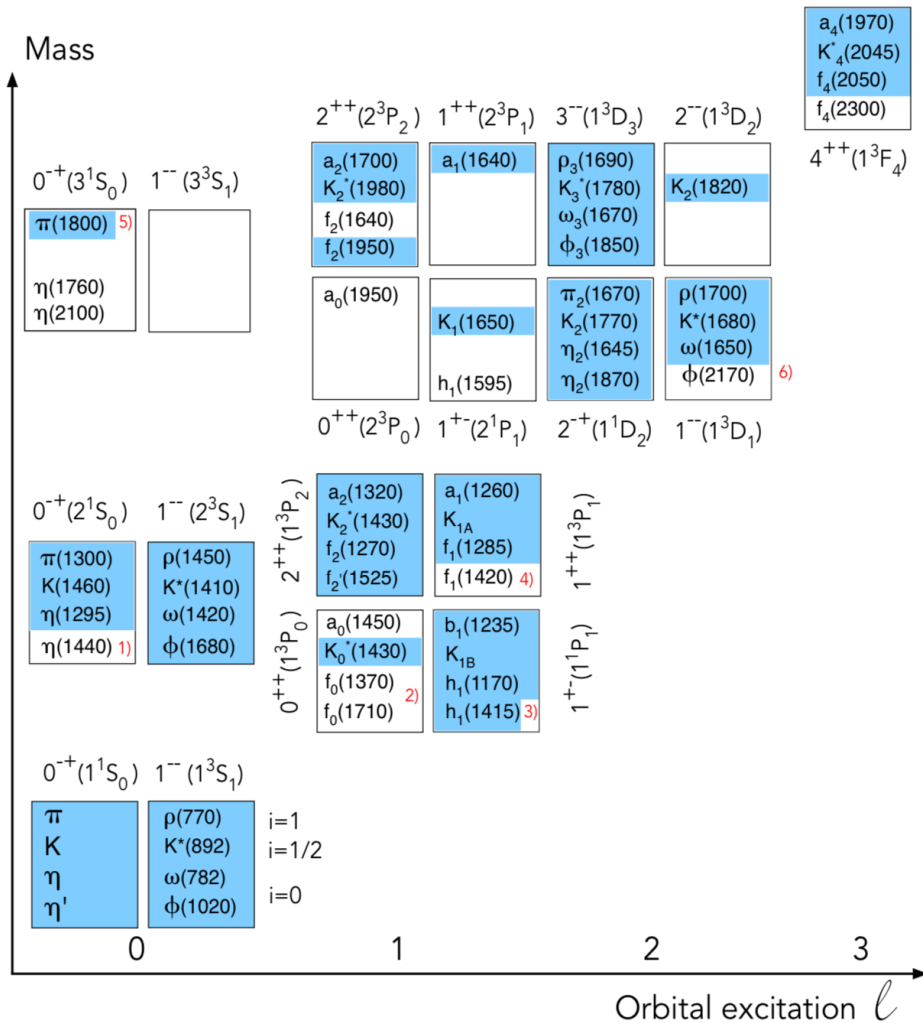
$$f' = \psi_8 \sin \theta + \psi_1 \cos \theta$$

$$C = (-1)^{L+S}$$

$$G = (-1)^{L+S+I}$$

for isospin multiplets

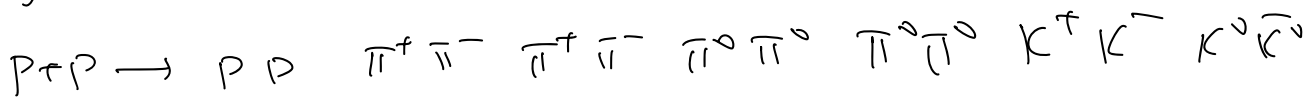
$$I = 1$$



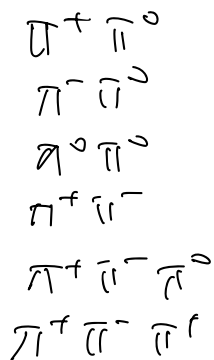
$$\rho^0 = |\bar{u}\bar{u} - d\bar{d}\rangle$$

$$p + p \rightarrow p + p + \rho^0$$

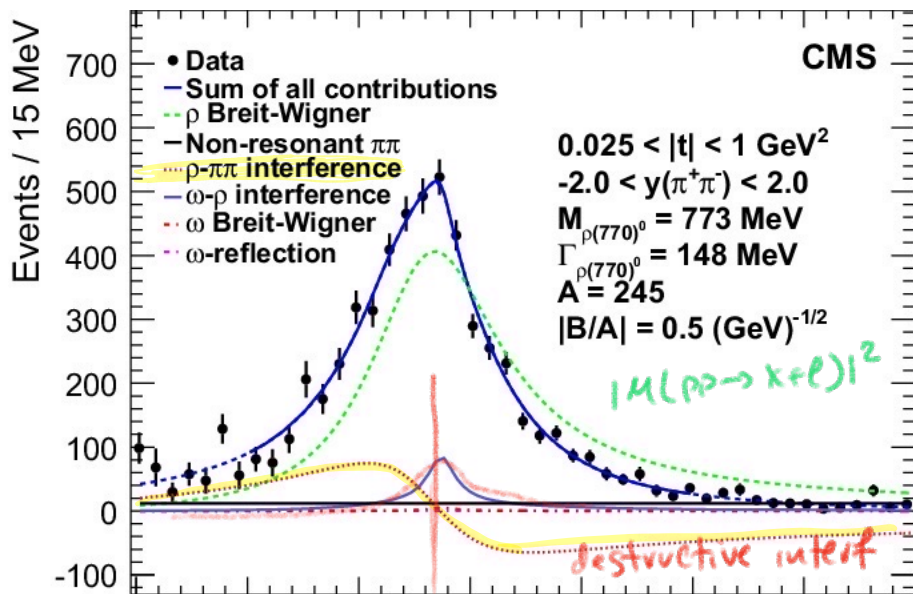
In general.



look at combinations of

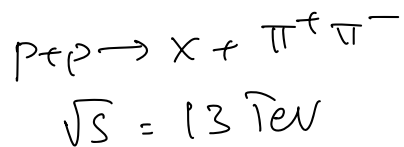


non resonant $\pi^+\pi^-$: $p\bar{p} \rightarrow p\bar{p}\pi^+\pi^-$



$$e = \frac{17}{2} - \frac{150}{770} < 1$$

$$> 0$$



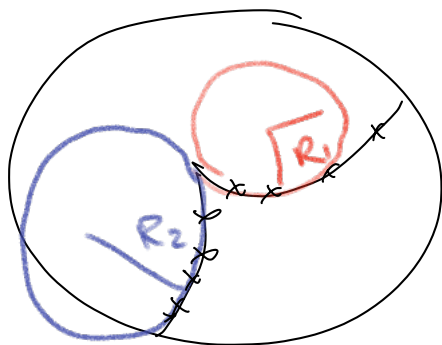
invariant mass $\pi^+ \bar{u}^-$

measure $|\vec{P}_{\pi^+}|$, $|\vec{P}_{\pi^-}|$

$$[GeV] \quad P = 0.3 \times B [T] \times R [m].$$

$$B = 3.8 \text{ T}$$

$$P_2 > P_1$$



$$M_{inv}^2 = (\underline{p}_1 + \underline{p}_2)^2 \quad \text{invariant mass.}$$

$$\#(\pi^+ \pi^-)_{\text{event } x} = \sigma(p_i \rightarrow \pi^+ \pi^-) \cdot \underbrace{\mathcal{L} \cdot \Delta t}_{137}$$

137 f_b^{-1}

2018-2022

$$p+p \rightarrow p+p + \pi^+ \pi^- \quad \text{non resonant}$$

$$\left. \begin{aligned} p+p &\leftarrow \rho^0 \rightarrow \pi^+ \pi^- = p+p + \pi^+ \pi^- \\ p+p &\leftarrow \omega \rightarrow \pi^+ \pi^- = p+p + \pi^+ \pi^- \end{aligned} \right\} \text{you only see } \pi^+ \pi^-$$

$$\sigma \propto |\mathcal{M}(pp \rightarrow x + \pi^+ \pi^-)|^2$$

$$\begin{aligned} \mathcal{M}(\pi^+ \pi^-) &= \mathcal{M}_1(pp \rightarrow x + \pi^+ \pi^- \text{ non resonant}) + \\ &\quad \mathcal{M}_2(pp \rightarrow x + \rho^0 \rightarrow x + \pi^+ \pi^-) + \\ &\quad \mathcal{M}_3(pp \rightarrow x + \omega \rightarrow x + \pi^+ \pi^-) \end{aligned}$$

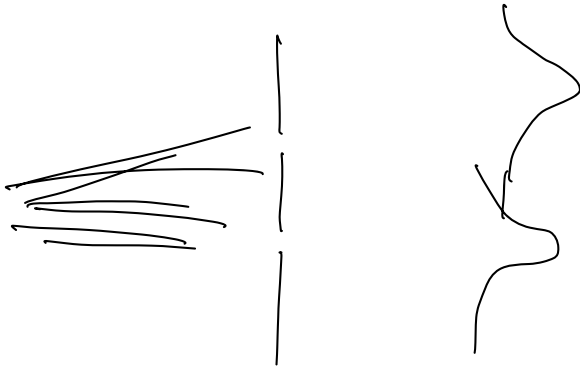
In general.

$$\begin{aligned} \mathcal{M}_1 &\neq \mathcal{M}_1^* \\ \mathcal{M}_2 &\neq \mathcal{M}_2^* \\ \mathcal{M}_3 &\neq \mathcal{M}_3^* \end{aligned}$$

$$|\mathcal{M}|^2 = |\mathcal{M}_1 + \mathcal{M}_2 + \mathcal{M}_3|^2 = |\mathcal{M}_1|^2 + |\mathcal{M}_2|^2 + |\mathcal{M}_3|^2 +$$

Interference terms.

$$\begin{aligned} &\mathcal{M}_1^\dagger \mathcal{M}_2 + \mathcal{M}_2^\dagger \mathcal{M}_1 + \mathcal{M}_1^\dagger \mathcal{M}_3 + \\ &\mathcal{M}_3^\dagger \mathcal{M}_1 + \mathcal{M}_2^\dagger \mathcal{M}_3 + \mathcal{M}_3^\dagger \mathcal{M}_2 \end{aligned}$$



Interference $< 0 \Rightarrow$ destructive interf.

$> 0 \Rightarrow$ constructive

$$A_1 = |A_1| e^{i\alpha_1} \quad A_2 = |A_2| e^{i\alpha_2} \quad A_3 = |A_3| e^{i\alpha_3}$$

$$|A_1 + A_2|^2 \geq |A_1|^2 + |A_2|^2 + \underbrace{A_1^\dagger A_2 + A_2^\dagger A_1}_{\geq 0}$$

depending on α_1, α_2

Decay of $\rho^0(770)$

	Mode	Fraction (Γ_i/Γ)	Scale factor/ Confidence level
Γ_1	$\pi^+\pi^-$	~ 100	%
Γ_2	K^+K^-		
$\rho(770)^\pm$ decays			
Γ_3	$\pi^\pm\pi^0$	~ 100	%
Γ_4	$\pi^\pm\gamma$	(4.5 ± 0.5)	$\times 10^{-4}$ S=2.2
Γ_5	$\pi^\pm\eta$	< 6	$\times 10^{-3}$ CL=84%
Γ_6	$\pi^\pm\pi^+\pi^-\pi^0$	< 2.0	$\times 10^{-3}$ CL=84%
$\rho(770)^0$ decays			
Γ_7	$\pi^+\pi^-$	~ 100	%
Γ_8	$\pi^+\pi^-\gamma$	(9.9 ± 1.6)	$\times 10^{-3}$
Γ_9	$\pi^0\gamma$	(4.7 ± 0.8)	$\times 10^{-4}$ S=1.7
Γ_{10}	$\eta\gamma$	(3.00 ± 0.21)	$\times 10^{-4}$
Γ_{11}	$\pi^0\pi^0\gamma$	(4.5 ± 0.8)	$\times 10^{-5}$
Γ_{12}	$\mu^+\mu^-$	[a] (4.55 ± 0.28)	$\times 10^{-5}$

<https://pdg.lbl.gov>

Page 11

Created: 8/11/2022 09:37

$$\rho^0 \rightarrow \pi^+\pi^-$$

$$\pi^0\pi^0$$

$$Q = m_\rho - 2m_\pi \approx 770 - 270 = 500 \text{ MeV}$$

$$\text{why } \text{Br}(\rho^0 \rightarrow \pi^0\pi^0) = 0$$

① $\mathbb{C}$ -Conjugation.

$$C_{\rho^0} = (-1)^{L+S} \Rightarrow C_{\rho^0} = (-1)^{1+0} = -1$$

vector meson $\Rightarrow S=1$.

ground state $\Rightarrow L=0$

$$\mathbb{C}(\pi^+\pi^-) = (-1)^{L+S} = (-1)^L$$

$$S: \pi^+\pi^-: \emptyset$$

$$L=1 \text{ possible} \Rightarrow \mathbb{C}(\pi^+\pi^-) = -1 \text{ allowed.}$$

$$\begin{array}{ccc} \rho^0 & \rightarrow & \pi^+\pi^- \\ J & 1 & S=0+L \Rightarrow L=1 \text{ from } J \text{ conservation.} \end{array}$$

$$\mathbb{C} \quad (-1) \quad (-1)$$

$$\begin{array}{ccccc} \text{Consider} & \rho^0 & \rightarrow & \pi^0 & \pi^0 \\ S & 1 & & 0 & 0 \\ L & 0 & & (1) & \\ J & 1 & & 1 & \end{array}$$

$$\mathbb{C}(\pi^0\pi^0) = (C_{\pi^0})^2 = +1$$

$$\rho^0 \rightarrow \pi\pi^0 \text{ violates } \mathbb{C}\text{-conservation.}$$

② $\pm 50 \text{ spin}$.

$$|\rho^0\rangle = |I=1, I_3=0\rangle$$

$$|\pi^0\rangle = |I=1, I_3=0\rangle$$

$$|\pi^0 \pi^0\rangle = 1 \otimes 1 = 0, 1, 2$$

$$|\pi^0 \pi^0\rangle = \alpha |I=2, I_3=0\rangle + \beta |I=1, I_3=0\rangle + \gamma |I=0, I_3=0\rangle$$

$$\alpha = \sqrt{\frac{2}{3}} \quad \beta = 0 \quad \gamma = -\sqrt{\frac{1}{3}}$$

$$|\pi^0 \pi^0\rangle = \sqrt{\frac{2}{3}} |2,0\rangle - \frac{1}{\sqrt{3}} |0,0\rangle$$

$$\mathcal{M}(\rho^0 \rightarrow \pi^0 \pi^0) = \langle 1,0 | H_{\text{strong}} \left(\sqrt{\frac{2}{3}} |2,0\rangle - \sqrt{\frac{1}{3}} |0,0\rangle \right)$$

$$H_{\text{strong}} \text{ conserves } I \Rightarrow \mathcal{M} = 0$$

Baryons in $SU(3)$ flavor

Baryon $\equiv q_1 q_2 q_3$

possible 27 states

$$B = 3 \otimes 3 \otimes 3 = 1 \oplus \underbrace{8_{\mu_{12}} \oplus 8_{\mu_{23}}}_{\text{Flavor Singlet}} \oplus 10_{\text{decuplet}}$$

Fundam.
repr.
of $SU(3)$

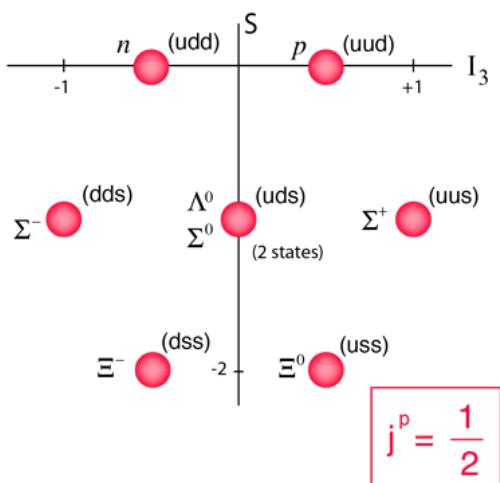
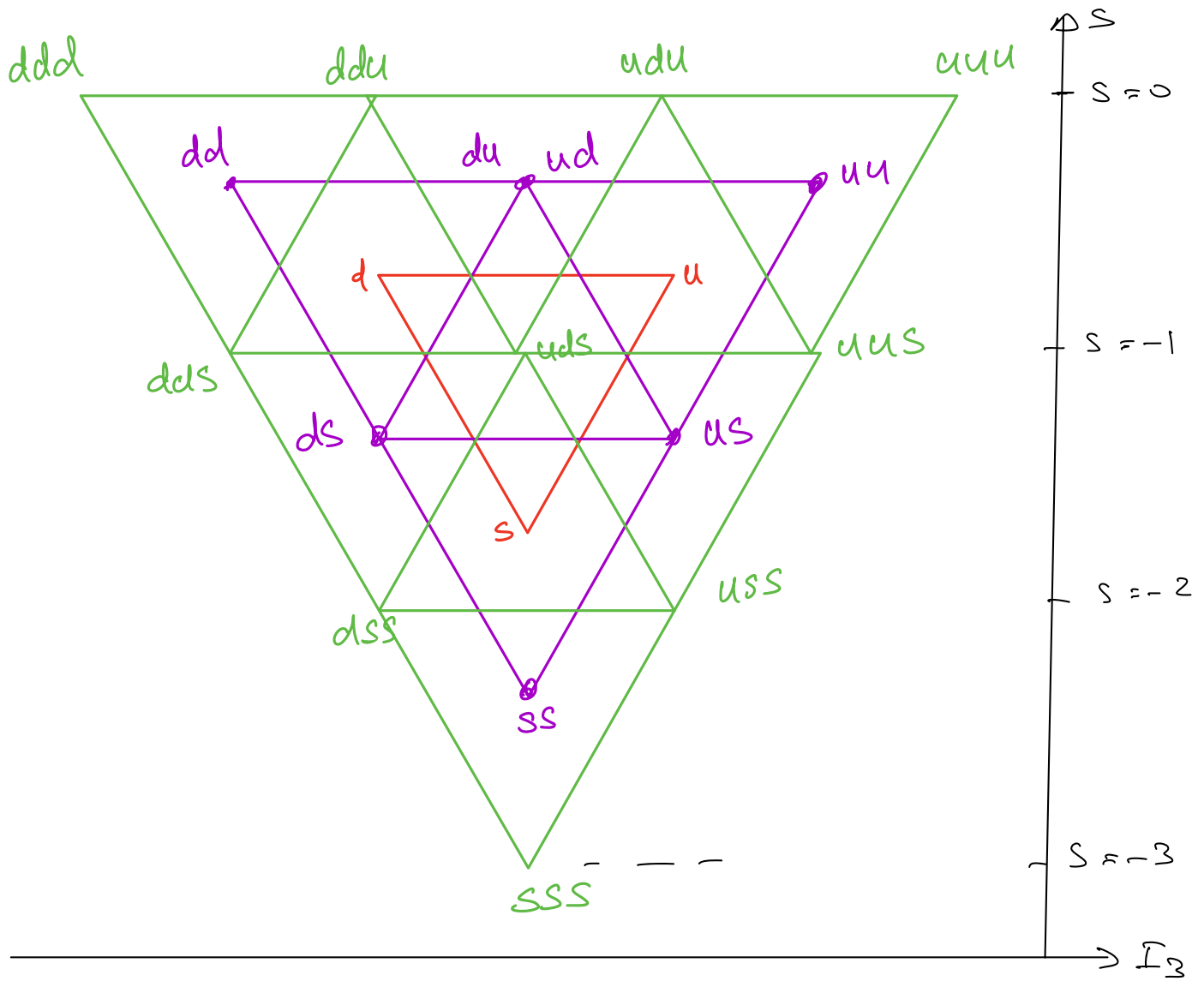
μ_{12} : symmetric under exchange $q_1 \leftrightarrow q_2$

μ_{23} symm. under $q_2 \leftrightarrow q_3$

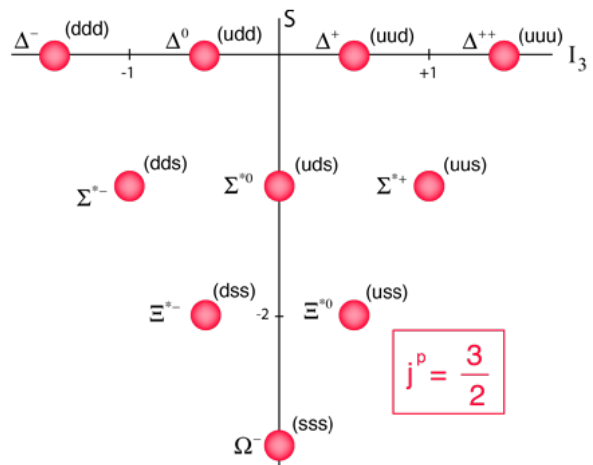
Experimentally we only see 8 + 10 physical states

1x1				+2 -1 1/15 1/5 3/5			
+1 +1 1 +1 +1				+1 0 8/15 1/6 -3/10			
+1 0 1/2 1/2				0 +1 2/5 -1/2 1/10			
0 +1 1/2 -1/2				+1 -1 1			
2 1 0				0 0 3			
0 0 0				-1 +1 1			
+1 -1 1/6 1/2 1/3				2 1			
0 0 2/3 0 -1/3				-1 -1			
-1 +1 1/6 -1/2 1/3				0 -1 1/2 1/2 2			
				-1 0 1/2 -1/2 -2			
				-1 -1 1			

$Y_\ell^{-m} = (-1)^m Y_\ell^{m*}$



physics states observed.



ground state $L=0$.

spin combinations: $\uparrow\uparrow\uparrow$ $S=3/2$
 $\uparrow\downarrow\uparrow$ $\uparrow\uparrow\downarrow$ $\downarrow\uparrow\uparrow$ $S=1/2$